



PYTHON NOTES — PYTHON DICTIONARIES AND TUPLES



TOPIC: DICTIONARIES (PART 3)

26. BEST PRACTICES WHEN USING DICTIONARIES

As my programs become larger, it is important to write dictionaries in a clear and organized way.

Use meaningful keys

Good:

```
student = {  
    'name': 'Christabel',  
    'age': 14  
}
```

Less clear:

```
student = {  
    'a': 'Christabel',  
    'b': 14  
}
```

Clear key names make programs easier to understand.

Keep key names consistent

If I begin with:

```
'title'
```

I should continue using:

```
'title'
```

and not suddenly switch to:

```
'Title'
```

or

```
'book_title'
```

unless there is a good reason.

Consistency makes programs easier to maintain.

Check before accessing

Instead of writing:

```
print(student['age'])
```

when I am unsure whether the key exists,

I should first write:

```
if 'age' in student:
```

This prevents errors.

27. COMMON MISTAKES

Mistake 1

Forgetting the colon.

Incorrect:

```
'name', 'Christabel'
```

Correct:

```
'name': 'Christabel'
```

Mistake 2

Forgetting commas between items.

Incorrect:

```
{  
'name': 'Chris'  
'age': 14  
}
```

Correct:

```
{  
'name': 'Chris',  
'age': 14  
}
```

Mistake 3

Using a key that does not exist.

Example:

```
student['height']
```

If the key is missing, Python raises an error.

Mistake 4

Using `.items()` incorrectly.

Incorrect:

```
for item in student.items():
```

This works, but the result is a tuple.

The cleaner approach is:

```
for key, value in student.items():
```

28. WHY DICTIONARIES ARE POWERFUL

Lists are excellent when the **position** of an item matters.

Example:

```
names[0]
```

Dictionaries are excellent when the **meaning** of the information matters.

Example:

```
student['name']
```

This makes programs much easier to read.

Compare:

```
student[0]
```

with:

```
student['favorite_subject']
```

The second version immediately tells us what information we are accessing.

29. REAL-WORLD USES OF DICTIONARIES

Python dictionaries appear almost everywhere in programming.

Examples include:

- Student Management Systems
- Library Management Systems
- Banking Applications
- Hospital Systems
- Contact Books
- Inventory Systems
- User Accounts
- Login Systems
- Shopping Websites
- Mobile Apps
- APIs
- Configuration Files
- JSON Data
- Databases

Learning dictionaries is therefore an important step toward writing real software.

30. WHAT I CAN DO NOW

After completing Dictionaries, I can:

- ☒ Create dictionaries
- ☒ Understand keys and values
- ☒ Access values using keys
- ☒ Add new key-value pairs
- ☒ Update existing values
- ☒ Remove information using `pop()`
- ☒ Check whether a key exists
- ☒ Use nested dictionaries
- ☒ Loop through dictionaries

✓ Use:

- `.items()`
- `.keys()`
- `.values()`

✓ Build programs using dictionaries

✓ Organize information more effectively than using many separate variables

MY PROGRESS

Current Topic: Dictionaries ✓

Concepts Learned

✓ What dictionaries are

✓ Creating dictionaries

✓ Keys

✓ Values

✓ Accessing values

✓ Adding items

✓ Updating items

✓ Removing items

✓ `pop()`

✓ Checking keys with `in`

✓ `.keys()`

✓ `.values()`

✓ `.items()`

✓ Nested dictionaries

✓ Looping through dictionaries

✓ Student Management System Project

DICTIONARIES COMPLETE

The Student Management System helped me combine many Python concepts into one project.

Through this project, I practiced:

- Functions
- Dictionaries
- Nested dictionaries
- Loops
- Conditions
- User input
- Updating data
- Adding data
- Removing data
- Displaying data
- Menu-driven programs

This project showed me that dictionaries are one of the most useful data structures in Python because they allow information to be stored in a meaningful and organized way.

NEXT TOPIC

The next part of these notes is:

PYTHON NOTES — PYTHON DICTIONARIES AND TUPLES (CONTINUED)

In the next note, I will learn about **Tuples**, including:

- What tuples are
- Packing and unpacking
- Indexing
- Slicing
- Looping through tuples
- Tuple methods
- Immutability

- Tuple vs List
 - Real-world uses of tuples
 - My Library Management System project
-

DICTIONARIES COMPLETE

I now understand how to use dictionaries to organize, update, retrieve, and manage structured information.

Dictionaries have become one of my strongest Python tools and will continue to appear throughout the rest of my Python journey.